Bhanuka Malith Silva PINCHAHEWAGE

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EXPERIENCE

AI Engineer (Data Science),
Future Secure AI, Sydney **April 2026 - present**

Domain: Financial Sector, Department: Macquarie Asset Management (MAM)

- Contributed as a data scientist within a cross-functional team to engineer a now in-production agentic AI framework consisting of complex document ingestion, business requirement validation and processing of unstructured / non-standardised financial instructions from clients.
- Owned and built accuracy critical core modules of the pipeline and integrated an automated end-to-end testing framework anchored by a golden dataset.
- Engaged in product strategy and architecture design discussions with senior engineers and stakeholders, helping shape the progressive versions of the product.

AI / ML Researcher,
Networked Systems and Security (NSS) Research Lab, Sydney **July 2022 – Mar 2026**

Domain: AI Engineering for Governance and Regulatory Compliance (GRC)

Lead Researcher – designed, alignment-trained, tested, at-scale deployed complex end-to-end AI frameworks and collaborated with legal experts:

- PrivPRISM: Engineered a hybrid encoder and decoder (LLM) framework to automate regulatory and platform compliance audits. Scaled processing to 1M+ Google Play app metadata records and conducted 10k+ app deep dives uncovering developer malpractices.
- PrivSTRUCT: Implemented a structure-aware LLM framework to untangle complex data purpose declarations from privacy policies achieving more than double the information recall performance against existing state-of-the-art tools. Conducted tests with over 3k policy documents.
- PrivDASH: Designed a deployment-ready privacy policy compliance analytics platform (LLM backend and React frontend) enabling interactive policy-risk visualisation for end-users and regulators.
- Entailment-verified LLMs: Designed explainable LLM pipelines for privacy policy interpretation with encoder-based verification, achieving over 90% agreement with legal expert annotations while reducing hallucinations or textual confusions.

Domain: AI Engineering for E-Safety and Network Security

- Designed a custom vision-language (VL) multimodal framework to identify content rating violations in popular app markets.
- Developed multimodal embedding fusion and large-scale counterpart search to detect mobile app metamorphosis (non-incremental changes such as re-births, re-branding, re-purposing etc.).
- TrafficLLM: Conducted LLM fine-tuning for structured feature extraction from network traffic.

Led and mentored graduate AI projects on;
- Detecting manipulative advertising practices in LLM outputs, and
- Assessing the privacy implications of generative AI based service provider disclosures.

Graduate Researcher | Lecturer - Autonomous Systems,
University of Moratuwa **March 2020 – June 2022**

- Supervised engineering projects on;
  (1) smart mobile receptionist robot with hardware constrained low-latency natural language (NL) conversation pipelines, and
  (2) collaborative ground-aerial systems for disaster response with DL frameworks for victim detection.
- Engineered remotely accessible simulation-first laboratory infrastructures for Autonomous Systems and Robotics with automated testing and access controls, enabling continuous learning for students during pandemic lockdowns.

Research Scholar – Cyber Physical Systems,
DATA61, CSIRO **Jun. – Dec. 2018 (on-site), Jan. 2019 – Jan. 2020 (remote)**

- Modelling electromechanical and thermodynamic subsystems of high-end (series elastic) actuators. Evaluated ML-based parameter estimation against real-world experiments.
- Developed a novel hybrid (leg-wheel) locomotion framework and accompanying gait algorithms. Concept was real world deployed using prototype hardware and full stack software development. A reconstructed digital elevation map (DEM) based on stereo vision was fed to a ML framework for dynamic adaptation of gait patterns.

SKILLS

GenAI, LLMs, LMMs (large multimodal models) – 4.5+ years

- Pipelines / Frameworks: Agentic workflows, LLMOps, RAG / Graph-RAG, Custom encoder+decoder (transformer) pipelines
- LLM Alignment: Supervised Fine-Tuning (SFT), Self-supervised FT, Direct Preference Optimisation (DPO), Model distillation
- Efficient Training: PEFT + QLoRA (bitsandbytes), Hardware acceleration (CUDA, MPS, AWS-Trainium)
- Inference / security: Prompt engineering and tuning, Prompt vulnerability testing (red teaming), Jailbreak defence (blue teaming)

Machine Learning (ML) & Computer Vision (CV) – 8+ years

- Architectures and techniques: CNNs, LSTMs, GANs
- Key Libraries/Frameworks: NumPy, Pandas, scikit-learn/image, OpenCV, TensorFlow, PyTorch
- Operations and security: MLOps, Adversarial Training

Programming Languages and Development

- Python, TypeScript, JavaScript, C/C++, MATLAB, React, Node.js

Cloud, Data and Infrastructure

- Cloud AI Platforms: Azure AI Foundry, Azure OpenAI, GCP Vertex AI, Hugging Face, AWS Bedrock, Textract, S3
- Data Systems: Vector Databases (Chroma, FAISS), Graph Databases (Neo4j), Relational (PostgreSQL), NoSQL (MongoDB)
- Orchestration, Tracking and Integrations: LangChain, Temporal SDK, Microsoft Graph, Weights & Biases (W&B), Grafna, Prometheus
- Developer Tools and Testing: Docker, Git, CI/CD, GitHub actions, Biome, Vitest

Technical leadership | Multi-disciplinary AI collaboration | Code review | Technical speaking & external representation | Responsible AI practices

EDUCATION

Ph.D., AI in Cybersecurity, School of Computer Science — The University of Sydney, Australia

Bachelor of the Science of Engineering – First Class Honours — University of Moratuwa, Sri Lanka

SELECTED TALKS AND PRESENTATIONS – AI FOR E-SAFETY

Mar. 2025	Breakout speaker – AISA Cyber Conference – "*Betrayed by Apps: Detecting e-Safety malpractices of mobile apps at-scale"* – Canberra
Nov. 2024	Breakout speaker – AISA Cyber Conference – "*Even LLMs can't understand privacy policies"* – Melbourne
Dec. 2024	Presenter – IEEE Conference on Building a Secure and Empowered Cyberspace – "*Entailment-driven privacy policy classification with LLMs"* – India

AWARDS AND HONOURS

Jul. 2022	University of Sydney International Student Award (SISA), Faculty of Engineering Research Stipend Scholarship and University of Sydney Tuition Fee Scholarship.
Sep. 2020	Migara Rathnatunga Trust Award 2019/2020 for undergraduate industrial training projects by Institution of Engineers Sri Lanka (IESL).
2015 – 2020	Dean's honour list appearances in undergraduate semesters 1,2,3,6,7 and 8 (University of Moratuwa).
2015 – 2020	Mahapola Higher Education Merit Scholarship by Ministry of Higher Education, Sri Lanka.
Feb. 2018	Dialog Sri Lanka - Merit Scholar.
2012 – 2017	14 awards in 165th, 164th and 162nd prize giving ceremonies of Royal College, Sri Lanka including: • J.N.Jinendradasa Memorial Prize for the **Best Performance** at the GCE A/L examination in Physical Science presented by Mr. P.T.Jinendradasa - 2014. • Asoka de Lanerolle Prize for the **Best Student** at the GCE A/L offering combined mathematics or biology, physics & chemistry presented by Mr. J.A.D. de Lanerolle - 2014. • Michael Gunaratne Memorial Prize for the **Best Performance** at GCE A/L examination (2014) entering the engineering faculty.
Nov. 2015	Sri Lanka Insurance Life - SUBA PATHUM National Scholarship Scheme awardee.
Dec. 2014	President Prize from Mithreepala Sirisena for the recognition of being **ranked second in national level** in GCE Advanced Level examination 2014, Sri Lanka.
2006	President Prize from Mahinda Rajapakshe for the performance in scholarship examination. (**District top 10** – 2006, Sri Lanka)

References available upon request